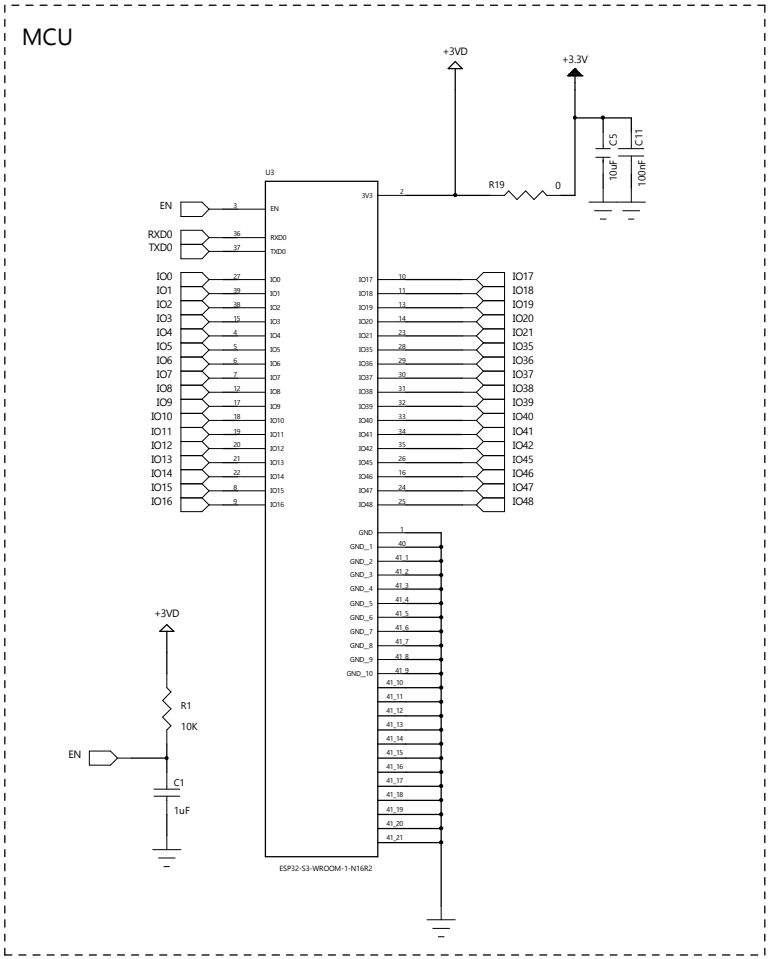
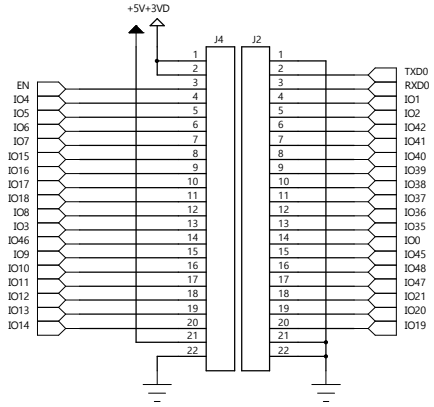
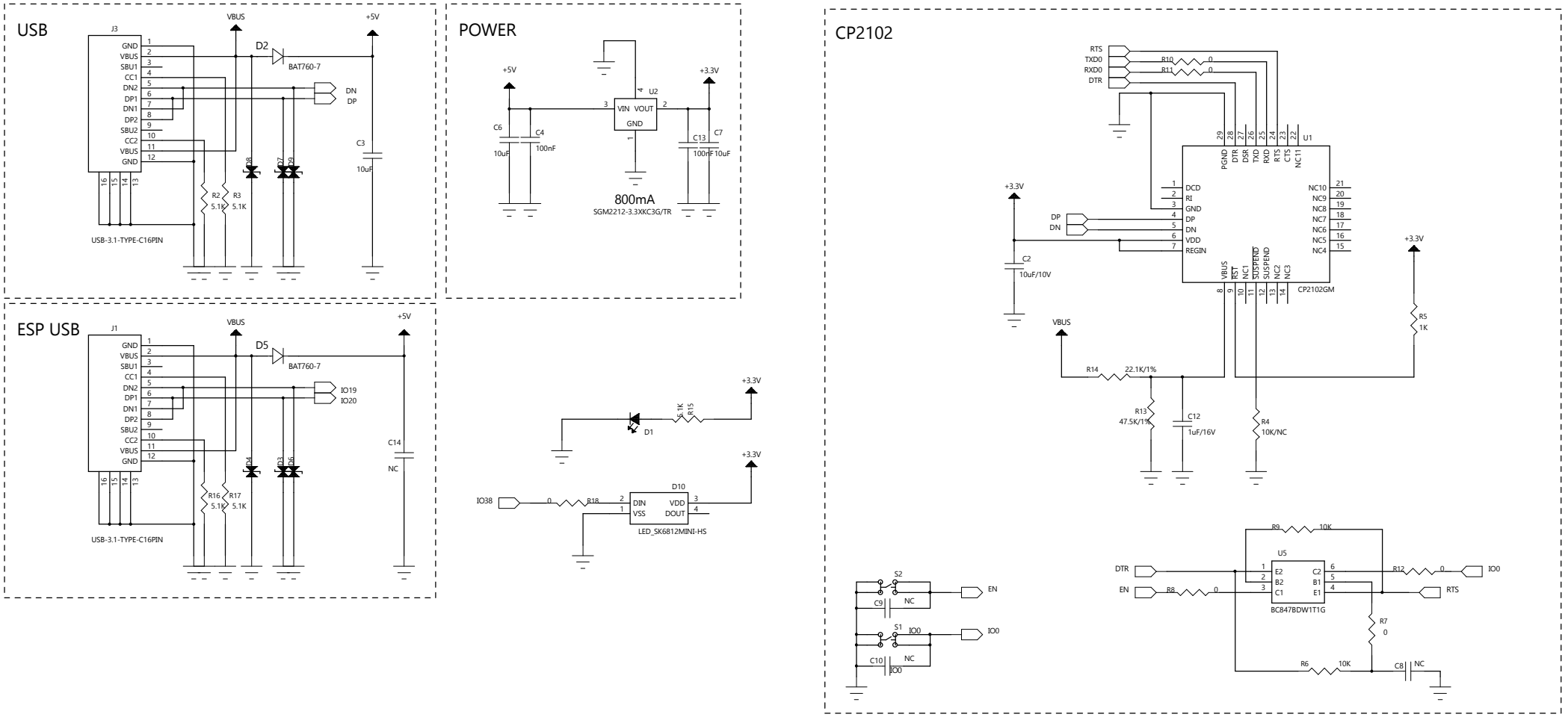

MCU
+3VD
+3.3V
C5
10uF
C11
100nF
U3
3V3
R19
0
EN
RXD0
TXD0
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO20
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
GND
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
41_1
41_2
41_3
41_4
41_5
41_6
41_7
41_8
41_9
41_10
41_11
41_12
41_13
41_14
41_15
41_16
41_17
41_18
41_19
41_20
41_21
+3VD
R1
10K
EN
C1
1uF
ESP32-S3-WROOM-1-N16R2


+5V+3VD
J4
J2
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
TXD0
RXD0
IO1
IO2
IO42
IO41
IO40
IO39
IO38
IO37
IO36
IO35
IO0
IO45
IO48
IO47
IO21
IO20
IO19


USB
J3
GND
VBUS
SBU1
CC1
DN2
DP1
DN1
DP2
SBU2
CC2
VBUS
GND
D2
BAT760-7
DN
DP
C3
10uF
R2
5.1K
R3
5.1K
USB-3.1-TYPE-C16PIN
POWER
U2
VIN
VOUT
GND
800mA
SGM2212-3.3XKC3G/TR
C6
10uF
C4
100nF
C13
100nF
C7
10uF
ESP USB
J1
D5
BAT760-7
IO19
IO20
C14
NC
R16
5.1K
R17
5.1K
USB-3.1-TYPE-C16PIN
D1
R15
1K
IO38
R18
D10
DIN
VDD
DOUT
VSS
LED_SK6812MINI-HS
CP2102
U1
RTS
TXD0
RXD0
DTR
R10
0
R11
0
DCD
RI
GND
DP
DN
VDD
REGIN
VBUS
RST
SUSPEND
SUSPEND
NC1
NC2
NC3
PGND
DTR
DSR
TXD
RXD
RTS
CTS
NC11
NC10
NC9
NC8
NC7
NC6
NC5
NC4
CP2102GM
C2
10uF/10V
R14
22.1K/1%
R13
47.5K/1%
C12
1uF/16V
R4
10K/NC
R5
1K
S2
C9
NC
EN
S1
IO0
C10
NC
IO0
R9
10K
U5
E2
B2
C1
C2
B1
E1
BC847BDW1T1G
R8
0
R12
0
R7
0
R6
10K
C8
NC